IOP Conference Series:
Earth and
Environmental Science

PURPOSE-LED PUBLISHING™

PAPER • OPEN ACCESS

Spatial patterns analysis of land use changes using spatial metrics in the peri-urban area of Surakarta City 2023

To cite this article: Rita Noviani *et al* 2024 *IOP Conf. Ser.: Earth Environ. Sci.* **1314** 012089

View the article online for updates and enhancements.

You may also like

- Modern Methods of Ensuring Sustainable Development of Rural Territories at the Regional Level
M S Yurkova, A I Firsov, V I Trofimova et al.
- Development Mode of Internet Economy Based on Artificial Intelligence Technology
Jiangnan He and Xiaoyin Yin
- The history and trends of semiconductor materials' development
Jitao Song

This content was downloaded from IP address 175.158.51.74 on 28/10/2026 at 13:05

Spatial patterns analysis of land use changes using spatial metrics in the peri-urban area of Surakarta City 2023

Rita Noviani[1*], Ahmad[1], and Istiyanti Nur Marfu'ah[1*]

[1] Geography Education Faculty of Teacher Training and Education

*Email: ritanoviani@staff.uns.ac.id, istiyantinm@student.uns.ac.id

Abstract. The development of cities and the transformation of suburbs are interrelated. Limited land in urban areas causes urban development to move outward towards peri-urban areas. Morphologically, the spatial development of urban fairy areas causes land use change. Surakarta City, as the mother city of the Subosukowonosraten Area, has experienced *over-capacity,* causing the effect of Surakarta City towards its periphery. This study aims to analyze spatial patterns of land use change using a spatial metrics approach. Research is quantitative descriptive research with a spatial approach. The results showed an increasing trend of residential land use change in all peri-urban zones of Surakarta City, with a faster rate of land use change in the peri-urban area of the inner zone as an area directly adjacent to the core city. The pattern of land use change shows the inner zone, and the outer zone shows the development of clustering patterns. In contrast, the urban shadow zone and rural hinterland zone show the development of longitudinal patterns.

1. Introduction

Land use activities are a significant issue and challenge for urban planning in realizing sustainable cities [1]. Land use analysis is important in understanding human activities' relationship with the environment [2]. Land use is important in maintaining and managing environmental quality [3]. The dynamics of land use change are strongly influenced by the increasing demand for space and the increasing population [4], which is not in line with the availability of fixed land [5].

Peri-urban areas have a high rate of land use change [6]. Peri-urban areas are characterized as areas with diverse and different land uses and functional areas with urban and rural sectors [7]. Urban sprawl phenomena often mark rapid land use change in suburban areas due to the expansion of the mother city into suburban areas [8][1]. The peri-urban area is an alternative to choosing where urban residents live because of the availability of large land [9]. With a low population density, peri-urban areas are the main destination to be used as a place to live for some residents due to the comfort level because it avoids noise compared to urban areas [10]. Bryant, 1982 classified peri-urban areas into four zones: inner zone, outer zone, urban shadow, and rural hinterland, which have different characteristics in each zone [7]. The inner zone is characterized by land that is experiencing a further transition from rural to urban properties and is identical to land that is being built. The outer zone is characterized by an area that is still dominated by a rural landscape but the urban nature is starting to look like single family housing. Urban shadow has areas with very minimal urban influence, but the influence of urban areas has begun to emerge, which can be seen from the travel patterns of part-time farmers and residents of small towns. The Rural Urban Fringe is synonymous with recreational use, extensive agricultural use, and open space.

Content from this work may be used under the terms of the Creative Commons Attribution 3.0 licence. Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.
Published under licence by IOP Publishing Ltd

Suburbanization in peri-urban areas of major cities in Indonesia continues to increase [9]. Surakarta City became the most populous city in Central Java in 2021 and has experienced overcapacity [11][12]. The limited availability of land in Surakarta City encourages the region's development outwards to form a new urban area in its periphery [13]. As a result, land use change in the peri-urban area of Surakarta City occurred very quickly.

Administratively, the peri-urban area of Surakarta City is located in three regencies, namely the Sukoharjo Regency, Boyolali Regency, and Karanganyar Regency. Judging from land use changes in 2003-2017, the development of Surakarta City tends to move southward, namely towards Sukoharjo Regency [14]. From 1999 to 2015, there was a shift in the direction of land fragmentation, which originally led to the northern area of Surakarta City to the south, in this case, Baki District, Grogol District, and around Karanganyar Regency [13]. This condition shows differences in spatial patterns of land use in the peri-urban area of Surakarta City, especially on the south and north sides of Surakarta City, with the east and west sides of Surakarta City, which get less influence. Analysis of spatial patterns as one of the important factors in the development of a city [15] is a very appropriate method for explaining land use changes in the peri-urban area of Surakarta City.

Spatial pattern analysis can be done by various methods, one of which is the spatial metric approach [15]. Spatial patterns concern a region's spatial characteristics and physical structure at any given time [16]. A spatial metric is an approach that can be used to formulate spatial patterns of a regional morphology based on matrix values [17]. Spatial metrics can analyze an area's structure, shape, and spatial pattern at the patch, class of patch, and landscape levels [18]. Spatial metrics can also measure land cover change and identify urban forms and urban sprawl phenomena [17]. Based on the explanation above, the spatial metrics approach is the right method for analyzing spatial patterns of land use change. This study uses a spatial metrics approach to analyze spatial patterns of land use change in the peri-urban area of Surakarta City in 2017-2023.

2. Methods

The research was conducted in the suburbs of Surakarta City The peri-urban area of Surakarta City has a total area of 244,938.94 hectares. In the study, the peri-urban area of Surakarta City was classified into four zones based on the classification by Bryant et al., 1982 consisting of the Inner Fringe Zone, Outer Fringe, Urban Shadow, and Rural Hinterland. The peri-urban area zone is presented in Figure 1. Figure 2 shows a map of the research site. Land use change in the peri-urban area of Surakarta City is land use in time series, namely 2017 and 2023, obtained from the interpretation of sentinel 2-A images. Sentinel 2-A images are processed using a likelihood-guided classification method, assessed as a classification method with a higher level of accuracy than ISODATA and k-means [18]. The image interpretation results are then carried out by field checking, observation, and interviews to verify land use and measure the accuracy and quality of image accuracy with the error matrix table and Kappa index. Using Sentinel 2-A imagery data, land use classification uses guided land use classification with likelihood methods processed with ArcGIS software. Land use is classified into five classes: settlements, industry, vegetation, agricultural land, and water bodies. The classification results are then verified by conducting field checking, observation, and interviews to test the accuracy level and quality level of accuracy using the error matrix table and Kappa Index. The processing phase carried out a land use change analysis by comparing land use in 2017 and 2023 with an overlay of interpreted land use maps. Analysis of spatial patterns of land use change with a matrix spatial approach using Fragstat software with the matrices used are LPI (Largest Patch Index), ED (Edge Density), GYRATE_AM (Area Weighted Mean Radius of Gyration), SHAPE_AM (Area Weighted Shape Index) FRAC_AM (Area Weighted Mean Fragmentation Index), PLADJ (Percentage of Like Adjacencies), IJI (Interspersion Juxtaposition Index), and COHESION (Cohesion Index).

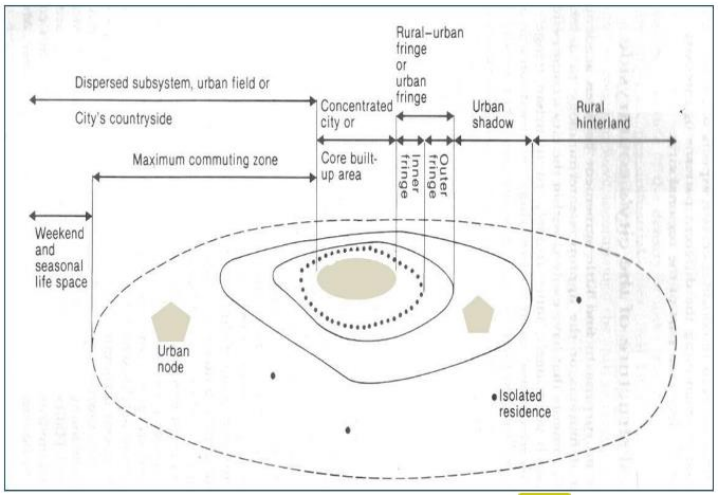


Figure 1. The Rural-Urban Fringe Scheme
Source: Thuo, 2013 (Modified from Bryant et.a.l, 1982)

3. Results and Discussion

3.1. Analysis of Land Use Change in the Peri-Urban Area of Surakarta City in 2017-2023

In general, the peri-urban area of Surakarta City is dominated by residential land use, industry, vegetation, and agricultural land. Land use classification in the peri-urban area of Surakarta City was carried out by processing Sentinel 2-A image data through a likelihood-guided classification method and obtained an accuracy rate of 86.88% with excellent accuracy quality based on the Kappa index.

Land use change in the peri-urban area of Surakarta City in 2017-2023 based on the results of processing land use raster data shows a development trend in residential and industrial land use, which tends to go east and south to west. In more detail, land use change in peri-urban areas is presented in Figure 2 and Table 1.

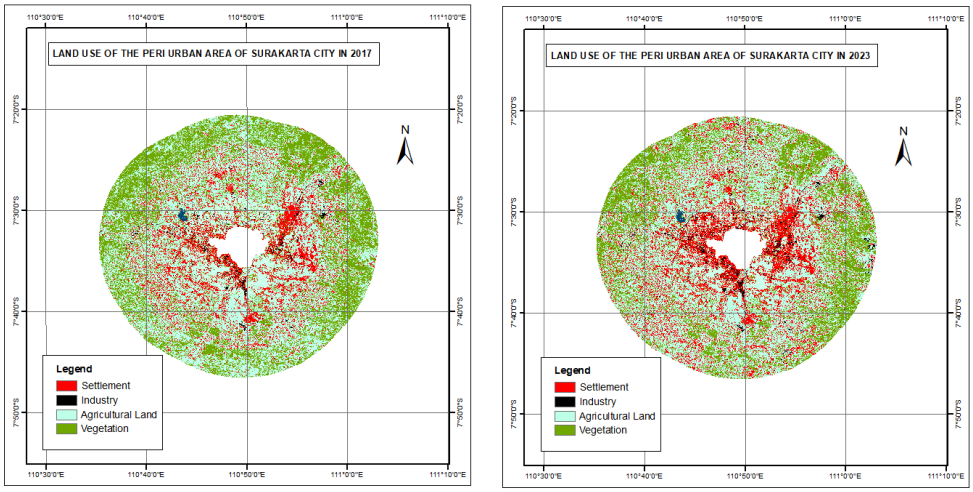


Figure 2. Land Use Map of The Peri-Urban Area of Surakarta City 2017 & 2023
Source: Reseach Analysis

Table 1. Land Use Change of The Peri-Urban Area of Surakarta City 2017-2023

Landuse	Pixel		Change	Hectare		Change
	2017	2023		2017	2023	
Settlement	318096	436039	117943	32592,19	44676,65	12,084.47
Industry	38463	63183	24720	3940,93	6473,74	2,532.82
Vegetation	1066286	1002605	-63681	109251,89	102727,13	-6,524.77
Agricultural Land	964586	885604	-78982	98831,69	90739,18	-8,092.51
Water Body	3022	3022	0	309,63	309,63	0,00
Total	2,390,453	2,390,453		244,938.94	244,938.94	

Source: Reseach Analysis

Table 1. Shows an increase in the area of built-up land use in the peri-urban area of Surakarta City, covering an area of 14,617.28 hectares (5.97%), namely residential land use increased by 12,084.47 hectares (2.64%) and industry covering an area of 2,532.82 hectares (1.03%) within six years. The largest reduction occurred in land use in agricultural land, covering an area of 8,092.51 hectares (3.30%). Vegetation land use from 2017-2023 has decreased by 6,524.77 hectares (2.66%). Fundamentally, increasing land use change in suburban areas occurs to meet the demands of good housing and quality of life along with the increasing population [19].

The high level of land use change on the south side of the peri-urban area of Surakarta City began with the development of The Solo Baru area in 2012 and further developed after 2016 [20][21]. The development of the built area on the south side is widening to the west after the construction of the primary arterial highway Solo-Yogyakarta, Solo-Semarang, Adi Sumarmo Highway and as a result of the construction of educational facilities [22]. Land use change on the eastern side of the peri-urban area of Surakarta City, especially in Jaten District, began in the 2000s due to its strategic location because it is on the border of Surakarta City to be the main alternative for site selection for both settlement and industry [23]. The development of the eastern zone accelerated after the construction of the Solo-Madiun and Solo-Tawangmangu highways was encouraged by the drainage permit of rice fields by the National Land Agency and settlement and industrial location permits by the Agency for the Application and Assessment of Technology in 2003-2009 [24][23]. In more detail, land use changes in each peri-urban zone in Surakarta City are as follows:

3.1.1. Inner Zone. The Inner Zone experienced a significant increase in built-up land, covering an area of 3,688.18 hectares (14.85% of the inner zone area). More complete land use changes in the inner zone are presented in Table 2.

Table 2. Land Use Change of Inner Zone of The Peri-Urban Area of Surakarta City 2017-2023

Landuse	Pixel		Change	Hectares		Change
	2017	2023		2017	2023	
Settlement	64,242	89,161	24,919	6,582.28	9,135.50	2,553.22
Industry	11,690	22,767	11,077	1,197.77	2,332.72	1,134.96
Vegetation	26,157	17,299	-8,858	2,680.06	1,772.47	-907.60
Agricultural Land	140,296	113,158	-27,138	14,374.83	11,594.25	-2,780.58
Total		242,385	242,385	24,834.94	24,834.94	

Source: Reseach Analysis

The inner zone is characterized as a transitional area from rural land use to urban land and is at the stage of the urban development process [7]. Based on Table 2, it is known that residential land use has increased the highest area, covering an area of 2,553.22 hectares or 10.28% of the inner zone area within six years. Industrial land use also increased in area by 1,134.96 hectares (4.28%). In line with the increase in built-up land, non-built-up land use has decreased in areas where agricultural land use has the largest reduction in the area of 2780.58 hectares (11.20%), and vegetation decreased by 907.60 hectares (3.65%) from the inner zone area of the peri-urban area of Surakarta City. The increase in residential land tends to move eastward and southward towards the west, as seen in Figure 3.

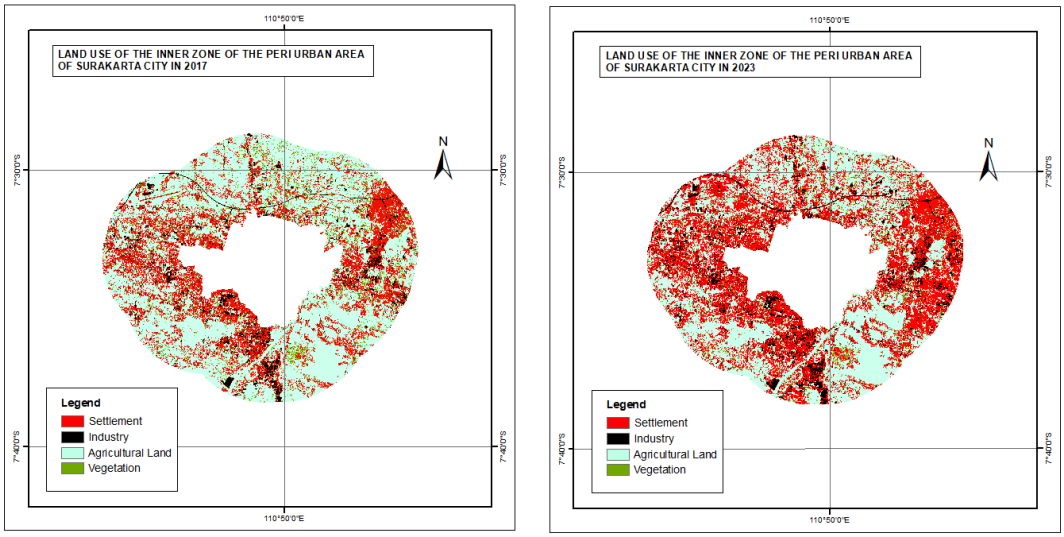


Figure 3. Land Use Map in the Inner Zone of the Peri-Urban Area of Surakarta City 2017 &; 2023
Source: Reseach Analysis

The eastward development is much influenced by the development of primary arterial roads and collector roads, namely the Solo-Madiun Road and the Solo-Tawangmangu Road, so it has high accessibility and increases the attractiveness of residents to live in the region [25]. The availability of large land at low prices, easy transportation to the city center, and the completeness of adequate social facilities also encourage the development of land use in settlements in the eastern zone of the peri-urban area of Surakarta City which is dominated by lower-middle class settlements inhabited by factory workers [26].

The development of settlements in the southern and western zones of the peri-urban area of Surakarta City is driven by the development of industrial estates and the availability of large land in Sukoharjo Regency and supported by easy accessibility on the south side of Surakarta City [14]. Population growth combined with the development of industrial estates, educational facilities, arterial highways, and the development of the JOGLOSEMAR Area (Yogyakarta-Solo-Semarang) encourages the development of residential areas in the peri-urban area of Surakarta City on the west side [27][22]. Recently, the development of settlements in suburban areas, especially the inner zone, tends to move towards clustered settlement patterns because the narrower land along the road encourages the development of widening settlements, filling the empty areas behind the established settlement areas [26].

3.1.2. Outer Zone. The outer zone is characterized as an area with an area still dominated by rural land use from the entire landscape. Still, there is penetration from urban areas in the form of the emergence of family housing that is already very clear [7]. Following the pattern of land use change in the inner zone, land use change in the peri-urban area of Surakarta City's outer zone tends to head east and south towards the west in the form of increased land-built settlements and industrial estates. More details of land use change in the outer zone can be seen in Table 3.

Table 3. Land Use Change in the Outer Zone in the Peri-Urban Area of Surakarta City in 2017-2023

Landuse	Pixel		Change	Hectares		Change	
	2017	2023		2017	2023	Ha	%
Settlement	76,752	117,596	40,844	7864.03	12,048.91	4,184.89	8.87
Industry	6,324	13,979	7,655	647.96	1,432.29	784.33	1.66
Vegetation	95,095	74,218	-20,877	9743.45	7,604.39	-2,139.06	
Agricultural Land	279,109	251,487	-27,622	28597.57	25,767.41	-2,830.16	-6.00
Water Body	3,022	3,022	0	30963	309.63	0.00	-4.54
Total	460,302	460,302		47,162.64	47,162.64		

Source: Reseach Analysis

Based on Table 3. It is known that residential land has experienced the largest area development, which is 8.87% (4,184.89 hectares) of the outer zone area. The development of industrial estates has increased by 784.33 hectares from the previous industrial area or 1.66% of the total area of the outer zone. Non-built-up land use continues to decrease from 2017-2023 in line with the increase in built-up land area. Agricultural land became non-developed land with the largest reduction in area of 2,830.16 hectares from the previous area (6% of the outer zone area). Vegetation decreased by 2,830.16 hectares or 4.54% of the total area of the outer zone.

The availability of large land in the outer zone and the condition of the area that began to be influenced by the inner zone became an alternative to the selection of settlements and industries in the outer zone [9]. The limited availability of living facilities such as housing, electricity, and water makes the periphery an alternative for choosing a place to live, one of which is in the outer zone [9]. The influence of the development of built land (settlements and industries) in the outer zone is still the same as the inner zone, which is driven by the high accessibility on the south side of Surakarta City, supported by the availability of land that is still extensive and the rapid development of industrial estates in Sukoharjo Regency encourages regional development, in this case, is built land tends to move towards the south which over time develops towards the west after the development of the triangle area Yogyakarta-Solo-Semarang (JOGLOSEMAR) [14][22]. The eastward development is influenced by high accessibility with the Solo-Madiun and Solo-Tawangmangu highways. It is supported by the availability of large land and the completeness of adequate social and transportation facilities [24][25][26]. More clearly, the development of built-up areas in the outer zone is presented in Figure 4.

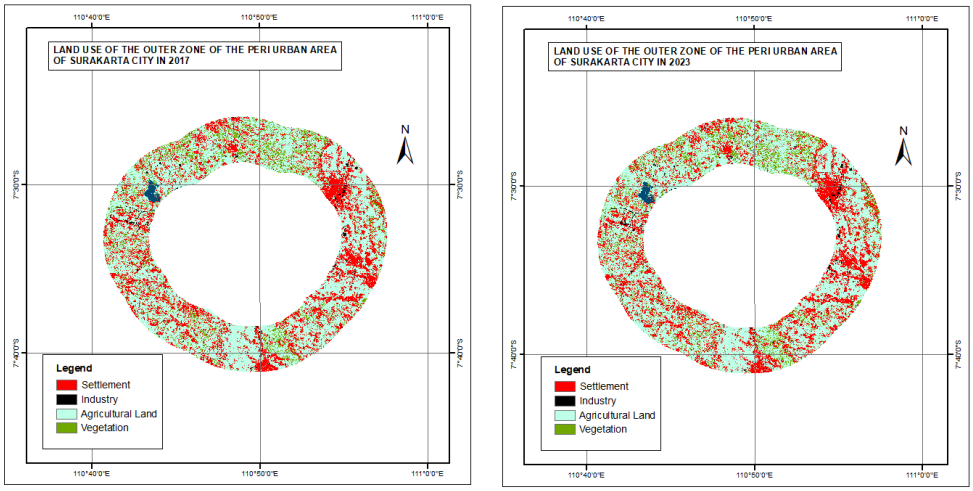


Figure 4. Land Use Map in the Outer Zone of Surakarta City Peri-Urban Area in 2017 &; 2023
Source: Reseach Analysis

3.1.3. Urban Shadow. Land use change in the urban shadow zone is still dominated by the development of settlements and industrial estates, which also encourage the reduction of agricultural land and vegetation. In more detail, land use change in urban shadow zones can be seen in Table 4.

Table 4. Land Use Change in Urban Shadow Zones 2017-2023

Landuse	Pixel			Hectares		Change	
	2017	2023		2017	2023	Ha	%
Settlement	87,883	110,425	22,542	9,004.60	11,314.28	2,309.68	3.30%
Industry	5,869	9,103	3,234	601.35	932.70	331.36	0.47%
Vegetation	376,679	361,637	-15,042	38,595.00	37,053.77	-1,541.22	-2.20%
Agricultural Land	212,013	201,279	-10,734	21,723.11	20,623.30	-1,099.82	-1.57%
Total	682,444	682,444		69,924.06	69,924.06		

Source: Reseach Analysis

Residential land use is still the dominant land use change in the urban shadow zone where in the 2017-2023 period, there is an increase of 2,309.68 hectares or 3.30% of the total peri-urban area of the urban shadow zone. Industrial development in the urban shadow zone is not as fast as in the inner and outer zones, which is only 0.47% of the area or 331.36 hectares. Vegetation is the land use with the highest reduction in area of 1,541.22 hectares (2.20%) and agricultural land use of 1,099.82 hectares (1.57%).

Unlike the case with the inner and outer zones, which still get a lot of influence from the city of Surakarta because of its location, which is still close to the core city, the urban shadow zone began to experience changes in the pattern of development of built areas which tend to spread patterns as can be seen in Figure 5.

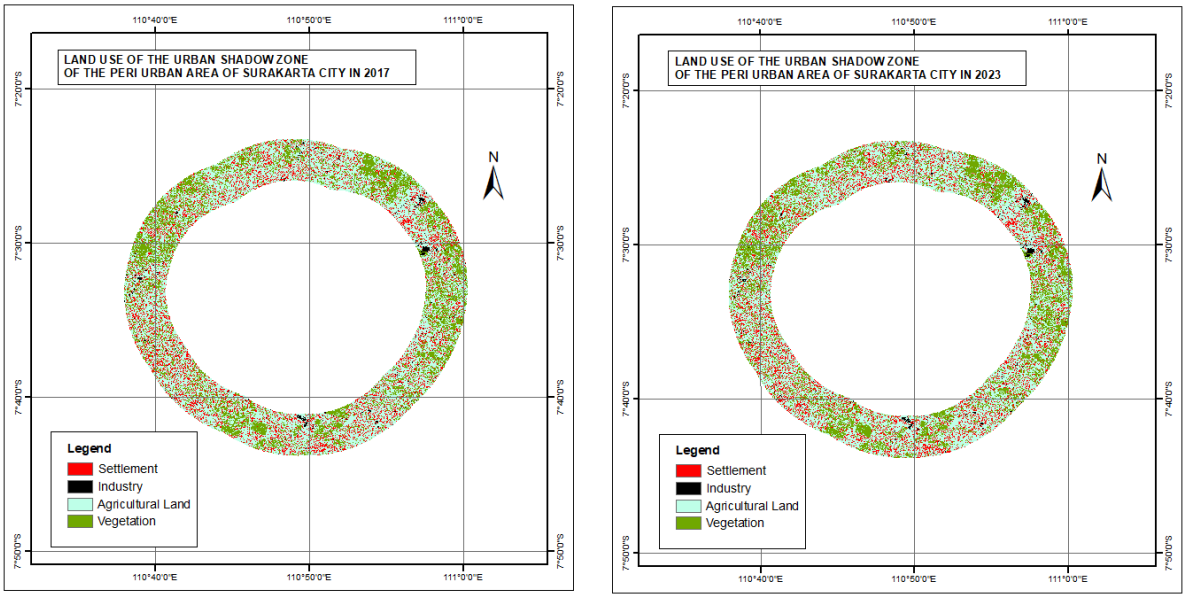


Figure 5. Land Use Map of Urban Shadow Zone of Peri-Urban Area Surakarta City in 2017 &; 2023
Source: Reseach Analysis

Figure 5 shows that the residential and industrial development pattern in the urban shadow zone is no longer much influenced by transportation networks such as the inner and outer zones. The urban shadow zone is characterized as an area with minimal urban influence. However, there is still an urban nature in the form of part-time peasant behavior, and residents are starting to be interested in urban culture [7]. The farther the distance between the region and the urban area, the pattern of settlement development is influenced by the physical conditions of the region, especially topography [28]. The urban shadow zone in the peri-urban area of Surakarta City is located in the foothills of the mountains, namely Mount Lawu on the east side and Mount Merapi and Mount Merbabu on the west side, causing the area to tend to be more diverse and high topography than the inner and outer zones which tend to have a flat topography but with slopes that are still gentle. The pattern of spreading settlements in urban shadow zones is also influenced by the number of residents who tend to be smaller, making settlements more likely to lead to a spreading pattern [29].

3.1.4. Rural Hinterland. The highest land use change in rural hinterland zones is still in the form of residential land use. In more detail, land use change in rural hinterland zones is presented in Table 5.

Table 5. Land Use Change in the Peri-Urban Hinterland Rural Zone of Surakarta City 2017-2023

Landuse	Pixel		Change	Hectares		Change	
	2017	2023		2017	2023	Ha	%
Settlement	89,219	118,857	29,638	9,142.45	12,179.51	3,037.06	2,95%
Industry	14,580	17,334	2,754	1,494.04	1,776.25	282.21	0,27%
Vegetation	568,355	549,451	-18,904	58,240.45	56,303.32	-1,937.13	-1,88%
Agricultural Land	333,168	319,680	-13,488	34,140.38	32,758.23	-1,382.14	-1,34%
Total	1,005,322	1,005,322		103,017.31	103,017.31		

Source: Reseach Analysis

Although not as rapid as in the inner and outer zones, the rural hinterland zone also experienced the largest increase in residential land use. In 2017-2023, there will be an increase in residential land use by 3,307.06 hectares or 2.95% of the total area. Industrial land use has also expanded by 282.21 hectares or only 0.27% of the total area. Similar to other zones, the rural hinterland zone experienced a reduction in non-developed land, with the largest reduction in the use of vegetation land covering an area of 1,937.13 hectares (1.88%) and agricultural land covering an area of 1,281.14 hectares (1.34%).

The urban shadow zone is characterized as an area with strong rural characteristics. It is dominated by large agricultural land and the availability of green open space and is generally used as a recreation area for urban residents [7]. The rural hinterland zone in the peri-urban area of Surakarta City is located in the mountainous area of Mount Lawu (east), Mount Merapi, and Mount Merbabu (west), as well as the urban shadow zone but with a higher location, topography, and slopes. With physical conditions in the form of mountainous areas with higher and steep topography, the development of settlements tends to spread in several areas and cluster around good tourist attractions natural attractions [29] such as natural attractions Grojogan Sewu waterfall, Cemoro Kandang, climbing Mount Lawu on the slopes of Mount Lawu; natural attractions Kedung Kayang Waterfall, Lempuyang Hill, Stabelan on the slopes of Mount Merapi; and Climbing Mount Merbabu. The pattern of land development developed in the rural hinterland zone is more clearly presented in Figure 6.

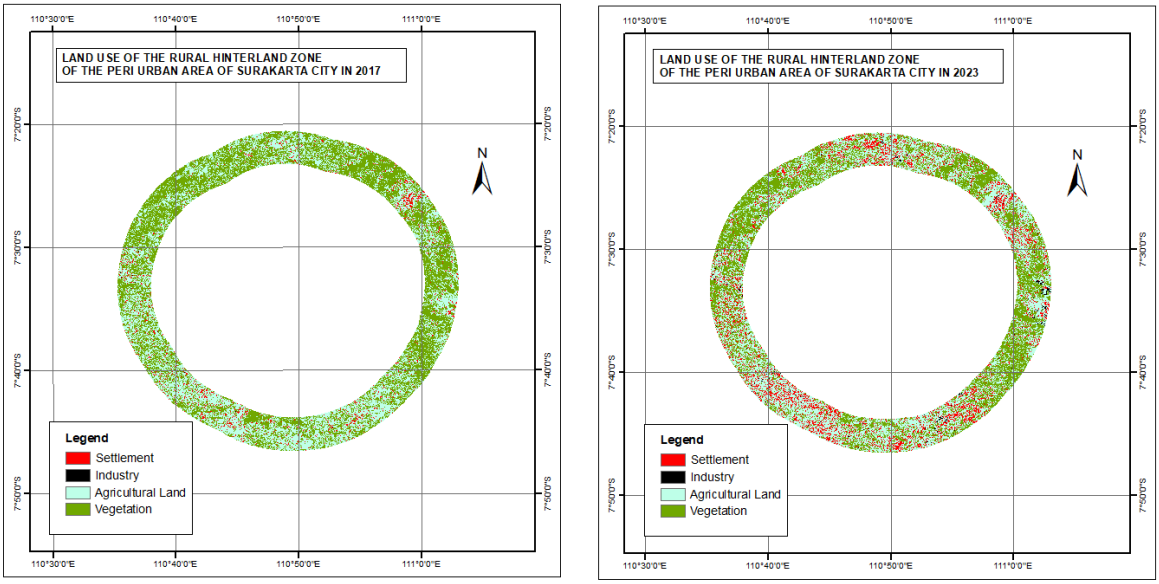


Figure 6. Land Use Map of Rural Hinterland Zone of Surakarta City Peri-Urban Area In 2017 & 2023
Source: Reseach Analysis

3.2. Spatial Patterns of Land Use Change in the Peri-Urban Area of Surakarta City with a Spatial Metric Approach

Analysis of land use change patterns using a spatial metric approach. The metrics used in spatial pattern analysis are LPI (Largest Patch Index), ED (Edge Density), GYRATE_AM (Area Weighted Mean Radius of Gyration), SHAPE_AM (Area Weighted Shape Index), FRAC_AM (Area Weighted Mean Fragmentation Index), PLADJ (Percentage of Like Adjacencies), IJI (Interspersion Juxtaposition Index), and COHESION (Cohesion Index) in more detail. The explanation of each metric is presented in Table 6.

Table 6. Spatial Pattern Analysis Metrics

No.	Metrics	Description
1.	LPI (Largest Patch Index)	Shows the largest landscape percentage of the total area (land use is dominant). The increasing value of LPI indicates the expansion of patches in dominant land uses
2.	ED (Edge Density)	Indicates the patch length directly adjacent to patches on other land uses. An increase in the ED value indicates an expansion of the patch, as seen from the increase in the number of pixels that intersect with pixels in other patches.
3.	GYRATE_AM (Area Weighted Mean Radius of Gyration)	Shows the distance between each cell and the center point of the patch, which is affected by the size and compression of the patch to measure how far the patch stretches from a landscape. The greater the GYRATE_AM value indicates the patch is getting elongated, so it will be close to whether the pattern of land use change is elongated or clustered.
4.	SHAPE_AM (Area weighted shape index)	Shows the complexity and irregularity of the shape of a patch. The higher patch value indicates the shape of a patch is more complex and irregular.

5.	FRAC_AM (Area Weighted Mean Fragmentation Index)	Indicates the fragmentation index of a class. A value of >1 in the metric FRAC_AM indicates a deviation in the Euclidean geometry that causes an increase in the complexity of the region's shape.
6.	PLADJ (Percentage of Like Adjacencies)	Indicates the continuity and density of a patch. A PLADJ metric with a value of 100 indicates that no pixel/area has different information from the dominant pixel/area. The higher the PLADJ value, the higher the continuity and density of patches
7.	IJI (Interspersion Juxtaposition Index)	Shows the proximity and diversity of the mix of adjacent patches. A high IJI metric value indicates a land use class with patches that are quite complex and often parallel to other patches of different class types. In contrast, a low IJI metric value indicates low patch complexity and distribution.
8.	COHESION (Cohesion Index)	Indicates the physical connectedness of patches within a class. The higher the increase in COHESION value, the higher the physical connectedness between patches in one type of class. A COHESION value close to 0 indicates an unconnected or fragmented patch. Meanwhile, the higher COHESION value indicates that the patches in one class are connected.

Source: Jatayu, 2020

3.2.1. Inner Zone. Based on spatial metric analysis using FRAGSTAT software, it is known that the measurement results are as presented in Table 7.

Table 7. Inner Zone FRAGSTAT Processing Results

Year	Metrics							
	LPI	ED	GYRATE_AM	SHAPE_AM	FRAC_AM	PLADJ	IJI	COHESION
2017	50.44	241.64	3,762.45	52.85	1.35	60.89	69.49	99.47
2023	27.39	187.83	3,153.45	28.27	1.34	69.50	72.95	99.13
	-23.05	-53.81	-609.00	-24.58	-0.01	8.61	3.46	-0.34

Source: Reseach Analysis

Based on Figure 3. It is known that the dominant land use in the inner zone is agricultural land. As the dominant land use, agricultural land experienced a decrease in LPI value of 23.05. The decline in the LPI value shows that agricultural land use has decreased widely and has begun to experience interference from land use development, especially settlements. In 2023, the ED value will decrease by 53.81, indicating that land use has increased in area and intervening in agricultural land use patches causes agricultural land to become narrower. In the inner zone, residential and industrial land use is the land use that has increased. Based on GYRATE_AM's known value from 2017-2023, it has decreased by 609, which shows that residential land use is developing to become more compact and clustered. From 2017-2023, the value of SHAPE_AM decreased by 24.58, indicating that the land use patch in the inner zone is growing more complex and irregular. The FRAC_AM value from 2017-2023 tends to stagnate and only decreases by 0.01. FRAC_AM values of 1.35 and 1.34 indicate the inner zone is experiencing shape complexity. PLADJ indicates the continuity and density of a patch. PLADJ values for 2017-2023 that do not reach 100 indicate that patches in the inner zone still have several pixels with different information in each patch. An increase in PLADJ value of 8.61 indicates patches in the inner zone have increased continuity and density. The IJI value has increased from 2017-2023 by 3.46, indicating that inner zone land use patches are increasingly complex. The COHESION value in 2017-2023 only decreased by 0.34. COHESION values of 99.47 in 2017 and 99.13 in 2023 indicate that the patch in each land use is physically connected. Based on this explanation, it can be seen that land use in the inner zone is developing increasingly complex but still physically connected. However, the use of

agricultural land as the largest land use in the inner zone began to experience intervention from the use of built-up land, namely settlements, and industry, causing the area of agricultural land to decrease.

3.2.2. Outer Zone. The results of measuring land use change with a spatial metric approach in the outer zone are more fully presented in Table 8.

Table 8. Inner Zone FRAGSTAT Processing Results

Year	Metrics							
	LPI	ED	GYRATE_AM	SHAPE_AM	FRAC_AM	PLADJ	IJI	COHESION
2017	74.36	164.56	9,092.84	64.94	1.38	73.26	49.05	99.75
2023	58.69	187.71	7,249.57	55.19	1.36	69.55	55.96	99.61
	-15.67	23.15	-1,843.26	-9.75	-0.02	-3.71	6.91	-0.14

Source: Reseach Analysis

Based on Table 8. It is known that the dominant land use in the outer zone, namely agricultural land, has also decreased in area, as seen from the LPI value, which has decreased from 2017-2023 by 15.67. The decline shows that there has been intervention from other land uses, namely settlements in the inner zone but not as large as the inner zone. The ED value in the outer zone has increased by 23.15 from 2017-2023, indicating an increase in land use area, which then takes over other land uses. In this case is the increase in built-up land use, which causes a decrease in non-built-up land use. The GYRATE_AM value from 2017-2023 decreased by 1,843.26, showing that land use is developing more and more clustered. An increase in the GYRATE_AM value in the outer zone greater than 1,237.26 from the inner zone indicates a greater grouping level in the outer zone region. The SHAPE_AM value for 2017-2023 in the outer zone decreased by 9.75, indicating that land use patches in the outer zone are more compact and regular, although not too significant. The FRAC_AM value in the outer zone also tends to stagnate with a reduction of only 0.02. The FRAC_AM value of 1.38 in 2017 and 1.36 in 2023 shows that the region's shape in the outer zone is increasingly complex because it has a value of more than 1. Unlike the inner zone, the value of PLADJ in the outer zone decreased by 3.71 from 2017-2023, which shows the continuity and density of land use patches are decreasing. From 2017-2023, the IJI value has increased by 6.91, which shows that the level of regional complexity is increasing. The increase in IJI value in the outer zone is greater than the inner zone of 3.45. Similar to the Inner zone, the level of connectivity between patches in the outer zone is high because the value is greater than 0 but has decreased, although not significantly at 0.14. Based on this explanation, it can be seen that the pattern of development in the outer zone still follows the pattern of development in the inner zone where, in general, the area develops more regularly. Still, the value of continuity, density, and connectedness between patches has decreased.

3.2.3. Urban Shadow. The results of measuring spatial patterns based on the metric approach in urban shadow zones are presented more fully in Table 9.

Table 9. FRAGSTAT Processing Results of Urban Shadow Zone

Year	Metrics							
	LPI	ED	GYRATE_AM	SHAPE_AM	FRAC_AM	PLADJ	IJI	COHESION
2017	55.90	258.50	1363.48	19.34	1.30	58.22	64.96	97.98
2023	5.84	237.85	9650.61	93.78	1.38	61.52	54.50	99.73
	50.06	-20.65	8287.14	74.44	0.08	3.31	-10.46	1.74

Source: Reseach Analysis

Based on Table 9, it is known that there are changes in patterns with inner and outer zones. Based on the LPI value, it is known that the urban shadow zone from 2017-2023 has decreased significantly by 50.06, which shows that the dominant land use, namely vegetation, has decreased in the area quite a lot from the previous year. The ED value in the urban shadow zone decreased by 20.65, indicating increased land use and intervening patches of vegetation land use as the dominant land use in this zone. In contrast to the inner and outer zones, urban shadow experienced an increase in GYRATE_AM value of 8,287.14, indicating that land use patches in this zone are experiencing development that tends to extend. An increase also occurred in the SHAPE_AM value of 74.44, which shows the level of land use complexity in the urban shadow zone is getting higher and more irregular in contrast to the inner and outer zones. FRAC_AM shows the fragmentation index of the region. In the urban shadow zone, the fragmentation value has increased by 0.08, which means that the level of fragmentation is getting higher, even if only slightly. The value of PLADJ in urban shadow zones has increased by 3.31, which means that the value of continuity and density of patches in each land use is increasing. The level of complexity in the urban shadow zone has decreased, as indicated by the decrease in IJI value by 10.46, which means that the area is getting more compact. The COHESION value for 2017-2023 has increased by 1.74, indicating that the connectedness between patches is increasing.

Based on the explanation above, it can be seen that the urban shadow zone has shown differences in development patterns. The urban shadow area tends to develop lengthwise. It is more compact than the inner and outer zones but is beginning to show the phenomenon of land fragmentation, although at very low levels.

3.2.4. Rural Hinterland. The spatial pattern of the calculation results with a more complete spatial metric approach is presented in Table 10.

Table 10. FRAGSTAT Processing Results of the Rural Hinterland Zone

Year	Metrics							
	LPI	ED	GYRATE_AM	SHAPE_AM	FRAC_AM	PLADJ	IJI	COHESION
2017	19.37	236.68	1855.35	19.61	1.29	61.71	79.03	98.17
2023	10.55	217.88	3956.49	42.67	1.36	64.72	52.67	99.30
	-8.82	-18.79	2101.14	23.06	0.06	3.01	-26.37	1.12

Source: Reseach Analysis

Based on the results of metric value calculations, it is known that the rural hinterland zone has a pattern of regional development that is almost the same as the urban shadow zone. Based on the LPI value, it is known that there was a reduction of 8.82, which shows that the dominant land use in general has decreased in area. The decrease in area aligns with the increase in built-up land, namely settlements and industry. The ED value in the rural hinterland zone decreased by 18.79, indicating increased land use and converting patches of vegetation land use as the dominant land use. The GYRATE_AM value has increased by 2,101.14, indicating that land use change tends to move lengthwise. The SHAPE_AM value increased by 23.06, indicating that the shape of the land use patch in the rural hinterland zone is increasingly complex and irregular. The FRAC_AM value in 2017 was 1.29 and increased by 0.6 to 1.36, indicating that this zone's fragmentation level has increased even slightly. The PLADJ value has increased by 3.01, indicating that this zone's continuity value and patch density value are increasing. The decrease in the IJI metric of 26.37 indicates that patches are increasingly complex. Like other zones, the rural hinterland zone experienced an increase in connectivity between patches by 1.12.

Based on the explanation above, it can be seen that the rural hinterland zone develops in clusters and the urban shadow zone. Still, the phenomenon of land fragmentation in this zone is greater than in the urban shadow zone, which is encouraged by the development of many natural attractions in this zone, increasing residents' attractiveness to start living in the zone.

4. Conclusions

In general, land use change occurs throughout the peri-urban area of Surakarta City but with different intensities and patterns. The most dominant change in use occurred in residential land use, which experienced the largest increase in area. Changes in residential land use in the inner zone experienced the largest increase in area compared to other zones. In line with the increase in residential land use, agricultural land use has experienced the greatest conversion compared to vegetation, while for land use, water bodies have a stagnant area.

Based on the results of the analysis using a spatial metric approach, it is known that the pattern of land use change in the inner and outer zones tends to form a clustering pattern caused by the development of built-up lands such as settlements and industries tend to fill the central regions due to the dense area along the highway. The urban shadow zone and rural hinterland show a pattern of longitudinal development based on the matrix value of the measurement results, although it has not occurred significantly.

References

[1] Bhat P A, Shafiq M U, Mir A A and Ahmed perves 2017 Urban sprawl and its impact on landuse/land cover dynamics of Dehradun City, India 513–21

[2] Das S and Sarkar R 2023 Spatiotemporal Change Detection Analysis of Land Cover and Land Use using Multi-Temporal Remote Sensed Imagery and Landscape Metrics: A Sub-Catchment Level Case Study of the Bhagirathi-Hugli River, West Bengal, India *IOP Conf. Ser. Earth Environ. Sci.* **1164**

[3] Maimaiti A, Wang L M, Yan F, Zhang J and Ma Y X 2017 Quantitative analysis of land use and land cover changes from the multi-temporal remote sensing data in the Bosten Lake Basin, Chinese Tian Shan *IOP Conf. Ser. Earth Environ. Sci.* **74**

[4] Rahmi K I N, Ali A, Maghribi A A, Aldiansyah S and Atiqi R 2022 Monitoring of land use land cover change using google earth engine in urban area: Kendari city 2000-2021 *IOP Conf. Ser. Earth Environ. Sci.* **950**

[5] Permatasari D and Pradoto W 2019 Analisis Perkembangan Kecamatan Colomadu Dan Kecamatan Grogol Sebagai Pklp Dan Wpu Surakarta Berdasarkan Faktor Internal *J. Pengemb. Kota* **7** 172–80

[6] Tavares A O, Pato R L and Magalhães M C 2012 Spatial and temporal land use change and occupation over the last half century in a peri-urban area *Appl. Geogr.* **34** 432–44

[7] Thuo A 2013 Unsettled Settled Spaces : Searching for a Theoretical *Int. J. Sci. Res. Publ.* **3**

[8] Kurnianingsih N A, Pratami M and Putri M B 2021 Karakteristik Pertumbuhan Penduduk Perdesaan pada Perkembangan Wilayah Peri-urban di Perbatasan Kota Surakarta *J. Sci. Appl. Technol.* **5** 138

[9] Cahya D L, Martini E and Kasikoen K M 2018 Urbanization and Land Use Changes in Peri-Urban Area using Spatial Analysis Methods (Case Study : Ciawi Urban Areas , Bogor Regency) Urbanization and Land Use Changes in Peri-Urban Area using Spatial Analysis Methods (Case Study : Ciawi Urban Areas , *IOP Conf. Ser. Earth Environ. Sci.* **123**

[10] Tagnan J N, Amponsah O, Takyi S A, Azunre G A and Braimah I 2022 A view of urban sprawl through the lens of family nuclearisation *Habitat Int.* **123** 102555

[11] Kusnandar V B 2022 Solo Kota Terpadat di Jawa Tengah pada 2021 *Databoks* 1

[12] Pasaribu I H and Mardiansjah F H 2017 Kajian Proses Perkembangan Pusat Pelayanan dan Pusat Permukiman di Kawasan Metropolitan Soloraya *Planologi* 1–19

[13] Mayca S N and Sri R G 2015 Analisis Fragmentasi Spasial Berbasis Citra Multitemporal Untuk Mengidentifikasi Fenomena Urban Sprawl Di Surakarta 1–10

[14] Fitriana A L, Subiyanto S and Firdaus H S 2017 Model Cellular Automata Markov untuk Prediksi Perkembangan Fisik Wilayah Permukiman Kota Surakarta Menggunakan Sistem Informasi Geografis *J. Geod. Undip* **6** 246–53

[15] Firmansyah F, Pamungkas A and Larasati K D 2018 Spatial pattern analysis using spatial metrics:

A case study in Surabaya, Indonesia *IOP Conf. Ser. Earth Environ. Sci.* **202**
[16] Andinasari C and Susetyo C 2019 Analisis Karakteristik Spasial Kawasan Permukiman dengan Spatial Metric di Distrik Jayapura Selatan, Kota Jayapura *J. Tek. ITS* **7**
[17] Hosea H, Pravitasari A E, Setiawan Y and Rustiadi E 2019 Landscape metric in the analysis of urban form in Cekungan Bandung urban region 72
[18] Jatayu A, Saizen I, Rustiadi E, Pribadi D O and Juanda B 2022 Urban Form Dynamics and Modelling towards Sustainable Hinterland Development in North Cianjur, Jakarta–Bandung Mega-Urban Region *Sustainability* **14** 907
[19] Septiani R, Citra I P A and Nugraha A S A 2019 Perbandingan Metode Supervised Classification dan Unsupervised Classification terhadap Penutup Lahan di Kabupaten Buleleng *J. Geogr. Media Inf. Pengemb. dan Profesi Kegeografian* **16** 90–6
[20] Prayitno G, Surjono, Hidayat A R T, Subagiyo A and Paramasasi N K 2018 Factors that effect to land use change in Pandaan District *IOP Conf. Ser. Earth Environ. Sci.* **202**
[21] Mataufani N A 2020 Pola Morfologi Kota Surakarta Dan Wilayah Peri Urban Di Sekitarnya 1–10
[22] Aulia F N 2018 Kajian Transformasi Spasial dan Dampaknya Terhadap Kualitas Permukiman di Kecamatan Grogol Kabupaten Sukoharjo Tahun 2006-2016 1–7
[23] Aini A N, Putri R A, Putri R A, Istanabi T and Istanabi T 2022 Kajian Pola Persebaran Permukiman Di Kecamatan Kartasura Kabupaten Sukoharjo *Desa-Kota* **4** 241
[24] Wibowo C S 2015 Dampak Pengalihan Fungsi Lahan Sawah Pada Produksi Padi Sampai Tahun 2018 Dan Implikasinya Terhadap Ketahanan Pangan Wilayah (Studi di Kecamatan Jaten Kabupaten Karanganyar Propinsi Jawa Tengah) *J. Ketahanan Nas.* **21** 107
[25] Fadlli Y I, Soedwiwahjono and Hardiana A 2016 Faktor yang Mempengaruhi Konversi Lahan Pertanian Studi Kasus : Kecamatan Jaten , Kabupaten Karanganyar *Arsitetkura* **4**
[26] Rindyantika P 2019 Analisis Spasial Perubahan Penggunaan Lahan Di Kecamatan Jaten Kabupaten Karanganyar Tahun 2008 dan 2018
[27] Oktavia P 2010 Proses Urbanisasi di Kecamatan Jaten Kabupaten Karanganyar
[28] Mardiansjah F H, Handayani W and Setyono J S 2018 Pertumbuhan Penduduk Perkotaan dan Perkembangan Pola Distribusinya pada Kawasan Metropolitan Surakarta *J. Wil. dan Lingkung.* **6** 215
[29] Karim H, Zhiddiq S and Suprapta 2017 UNM Geographic Journal *UNM Geogr. J.* **1** 48–54